



2025

H23(B)

BOOKLET NO.

207066

Botany

Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

**SECTION – A**

Q1. Answer the following in about **150** words. **(10×5=50)**

- Q1. (a)** Compare the major plant immune signalling pathways involved on pathogen recognition and defence activation. Discuss recent scientific insights into the integration and crosstalk between receptor mediated Pattern-Triggered Immunity (PTI) and intracellular Effector-Triggered Immunity (ETI). Support your answer with examples of receptor types and signalling molecules illustrating how these pathways cooperate to enhance plant Immunity. **10**
- Q1. (b)** Describe the historical development of genetics, highlighting key discoveries that shaped the gene and allele concepts. Explain the difference between a gene and an allele and discuss recent advancements that have expanded our understanding of these concepts. **10**
- Q1. (c)** Critically analyze the methods of plant breeding, focusing on selection and hybridization techniques such as pedigree method, back cross method, mass selection and bulk method. Discuss their principles, advantages, limitations and applications in crop improvement. **10**
- Q1. (d)** Describe the structure and functions of plant cell membranes with special emphasis on cell adhesion, type of membrane transport and the role of vesicular transport. Explain how these processes contribute to maintaining homeostasis and intracellular communication in plants. **10**
- Q1. (e)** Describe the structure and major components of plant extracellular matrix (cell wall). Discuss its functions in plant growth, defence and intracellular communication. Highlight recent insights into cell wall remodelling and how understanding its dynamic nature can be applied in improving plant resistance and development. **10**
-



- Q2. (a)** Explain the molecular regulation of the plant cell cycle, focusing on key regulatory proteins such as Cyclin-Dependent-Kinases (CDKs) and cyclins. Discuss recent insights into how plants integrate cell cycle progression with developmental signals and environmental stress responses. **20**
- Q2. (b)** Explain linkage and crossing over in plants. Describe gene mapping and the use of molecular maps, including their significance in plant genetics and breeding. **15**
- Q2. (c)** Explain the concepts of standard deviation and coefficient of variation. Discuss their importance and applications in plant sciences. Solve the following numerical problem to calculate the standard deviation for given sample data of plant heights.
- Given heights (cm) of 20 plants :10, 11, 9.5, 10, 11, 10, 11.5, 11, 10.5, 11.5, 9, 10, 10.5, 11.5, 11, 9.5, 10.5, 11, 11, 10.5.
- (Calculate the sample standard deviation rounded to two decimal places.) **15**
-
- Q3. (a)** Explain the genetic code and the regulation of gene expression in plants. Discuss the mechanisms and significance of gene silencing and describe multigene families with focus on recent insights and applications in plant systems. **20**
- Q3. (b)** Explain numerical and structural variations in plant chromosomes. Discuss their significance in plant evolution, adaptation and breeding with insights and applications. **15**
- Q3. (c)** Explain the development and use of molecular markers in plant breeding. Discuss recent advances and provide examples illustrating their applications in crop improvement. **15**
-



- Q4. (a)** Explain the methods of gene transfer used in genetic engineering of plants. Discuss the developments and significance of transgenic crops, with emphasis on biosafety considerations. Critically analyse the reservation and regulatory framework surrounding the use of genetically modified crops in India. **20**
- Q4. (b)** Describe the detailed structure of chloroplasts and their essential functions in plants. Discuss variations in chloroplast structure and recent insights into their roles in metabolism, stress response and agricultural applications. **15**
- Q4. (c)** Explain the cytoplasmic inheritance and the role of cytoplasmic genes in plants. Discuss examples, mechanisms and advances in understanding their significance in plant genetics. **15**

SECTION – B

- Q5.** Answer the following in about **150** words. **(10×5=50)**
- Q5. (a)** Describe the physiological processes governing fruit and seed development, explaining the roles of hormones and environmental factors. Discuss how understanding seed dormancy and germination mechanisms contributes to improved seed storage, viability maintenance and crop production under changing climatic conditions. **10**
- Q5. (b)** Describe the two main stages of photosynthesis. How C₃, C₄ and CAM plants differ in the way they fix carbon dioxide ? Which of these plants is best suited for hot and dry environments and why ? **10**
-



- Q5. (c)** Define the concept of sustainable development. Discuss its importance and significance in balancing economic growth, social inclusion and environmental protection for present and future generations. **10**
- Q5. (d)** Define plant succession and explain the main stages involved in this ecological process. Differentiate between primary and secondary succession with suitable examples. Discuss the ecological significance of plant succession in ecosystem development and restoration. **10**
- Q5. (e)** Explain the Chemiosmotic theory and describe how it accounts for ATP synthesis in mitochondria and chloroplasts. Discuss the role of the proton gradient and ATP synthase in this process. Also, evaluate one recent insight or advancement that has deepened our understanding or enhanced the efficiency of ATP production based on this theory. **10**
-
- Q6. (a)** Explain the concept and fundamentals of biological nitrogen fixation and nitrogen metabolism, detailing the role of the nitrogenase enzyme and energy requirements. Discuss the different types of nitrogen-fixing organisms and their symbiotic or free-living lifestyles. Additionally, describe one recent insight or application derived from the understanding of nitrogen fixation that has practical significance in agriculture or industry. **20**
- Q6. (b)** Explain the concept of photoperiodism and its role in regulating flowering in plants. Describe the molecular mechanisms by which plants detect daylength and induce flowering. How can this knowledge be applied to agricultural practices to improve crop yield? **15**
-



- Q6. (c)** Describe the major Phytogeographical regions of India, highlighting their distinctive climatic and vegetation characteristics. Discuss the importance and ecological significance of these regions in terms of biodiversity conservation and sustainable resource management. **15**
-
- Q7. (a)** Describe the major forest types of India based on climatic and geographical factors. Discuss the distinctive characteristics, distribution and ecological significance of each forest type. How do these forests contribute to biodiversity conservation and sustainable resource management in India? **20**
- Q7. (b)** Explain the importance of secondary metabolites in plants. Discuss their roles in plant defence, reproduction and environmental adaptation. Highlight the significance of these compounds in medicine, agriculture and industry. **15**
- Q7. (c)** Define the concept of endangered plants and endemism. Explain the role of the IUCN Red list and the Red Data Book in the conservation of plant biodiversity. Discuss the importance of these tools in identifying and protecting endangered plant species, with examples from India. **15**
-
- Q8. (a)** Explain the physiological processes underlying fruit ripening and describe the molecular mechanisms controlling this complex developmental stage. Discuss the role of key hormones such as ethylene and abscisic acid in regulating ripening. How can manipulation of these molecular pathways can be applied in horticulture to improve fruit quality, shelf-life and post harvest performance? **20**
-



- Q8. (b)** Describe the mechanism of phloem transport (translocation) in plants. Explain how sugars are loaded into the phloem at the source and unloaded at the sink. Discuss the role of the pressure – flow hypothesis in explaining phloem transport. Include the energy requirements and factor affecting this process. **15**
- Q8. (c)** Explain the concepts of global warming and climate change. Discuss the role of deforestation in accelerating these environmental issues. How does deforestation contribute to green house gas emissions and disrupt the carbon cycle ? Suggest measures to reduce the impact of deforestation on climate change. **15**
-

*
AL*
*